Grade 11 Term 2 Home Practice – C1

Parameters and sample statistics

Global Economics – 2026–2027

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Evaluated criteria

C1 Distinguishes population parameters from sample statistics by interpreting and comparing the notation for population and sample means, standard deviations, and proportions in economic and financial contexts.

How to practise. Try each task before opening its blue solution link. The arrows return to the contents, problem list or current problem.

1 Worked Solution Examples

Problems:

1.1. A market association's sales survey (C1)
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1.1 A market association's sales survey (C1)

A market association wants to describe all 1,200 stalls licensed for June. From 60 randomly selected stalls, it finds a mean June sales revenue of 4.8 million COP and a sample standard deviation of 1.2 million COP. Eighteen sampled stalls accept digital payment.

Questions/Tasks

C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 To check how the survey results vary across samples, the association draws another random sample from the same June list. It finds a mean of 5.1 million COP, a standard deviation of 1.0 million COP and a digital-payment share of 0.25. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

Sample (green)	**Population (red)**
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)
Sample proportion ($\hat{p}$)	Population proportion (p)
Sample (selected units)	Population (all units of interest)

C1 – Match each quantity to its symbol

The population is all 1,200 June-licensed stalls. The sample is the 60 selected stalls. Each pair describes the same feature at a different scale.

Population parameter	Sample statistic
μ, mean June sales across all licensed stalls	$\bar{x} = 4.8$ million COP, mean across the 60
σ, SD of June sales across all licensed stalls	$s = 1.2$ million COP, SD across the 60
p, fraction of licensed stalls accepting digital payment	$\hat{p} = 18/60 = 0.30$, fraction among the 60

The two sales measures have units of million COP. The payment proportions are unitless and can also be written as percentages.



C1 – Compare what changes across samples

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All 1,200 June-licensed stalls	Same June list	μ, σ and p stay fixed.
Sample	First 60 selected stalls	Second random sample from the June list	A new draw can produce different statistics.
Mean	$\bar{x} = 4.8$ million COP	$\bar{x} = 5.1$ million COP	$5.1 - 4.8 = +0.3$ million COP in $\bar{x}$.
Standard deviation	$s = 1.2$ million COP	$s = 1.0$ million COP	$1.0 - 1.2 = -0.2$ million COP in s.
Digital-payment share	$\hat{p} = 18/60 = 0.30$	$\hat{p} = 0.25$	$0.25 - 0.30 = -0.05$ in $\hat{p}$ (-5 percentage points).



1.2 A supermarket's checkout survey (C1)

A supermarket manager wants to describe all 1,500 purchases made at one store on Saturday. In a random sample of 50 receipts, the mean purchase value is 64 thousand COP and the sample standard deviation is 18 thousand COP. Twenty of those purchases used a loyalty card.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 Before reporting the results, the manager asks a colleague to draw another random sample of 50 receipts from that store's Saturday records. The new sample has a mean purchase value of 61 thousand COP and a standard deviation of 20 thousand COP. Twenty-five of these purchases used a loyalty card. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The population is all 1,500 Saturday purchases at the store. The sample is the 50 purchases represented by the selected receipts.

Population parameter	Sample statistic
μ, mean value of all Saturday purchases	$\bar{x} = 64$ thousand COP, mean among the 50 purchases
σ, standard deviation of value across all Saturday purchases	$s = 18$ thousand COP, standard deviation among the 50 purchases
p, fraction of all Saturday purchases that used a loyalty card	$\hat{p} = 20/50 = 0.40$, fraction among the 50 purchases



C1 – Compare what changes across samples

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All 1,500 Saturday purchases at the store	Same 1,500 purchases	μ, σ and p stay fixed.
Sample	First draw of 50 receipts	Second draw of 50 receipts	The sample size stays the same. The draw is new.
Mean	$\bar{x} = 64$ thousand COP	$\bar{x} = 61$ thousand COP	$61 - 64 = -3$ thousand COP in $\bar{x}$.
Standard deviation	$s = 18$ thousand COP	$s = 20$ thousand COP	$20 - 18 = +2$ thousand COP in s.
Loyalty-card share	$\hat{p} = 20/50 = 0.40$	$\hat{p} = 25/50 = 0.50$	$0.50 - 0.40 = +0.10$ in $\hat{p}$ ($+10$ percentage points).



1.3 A transit agency's fare review (C1)

A transit agency wants to describe all 8,400 bus boardings on its central routes last Monday. In a random sample of 100 boarding records, the mean fare paid is 3.2 thousand COP and the sample standard deviation is 0.8 thousand COP. Thirty-five of the sampled boardings used a discounted fare.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 For a fare review, the agency's director asks to include suburban routes alongside the central routes, using the same Monday records. The combined routes have 12,000 boardings. A random sample of 100 from the combined records gives a mean fare of 3.4 thousand COP and a standard deviation of 0.9 thousand COP. Thirty sampled boardings used a discounted fare. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The population is all 8,400 Monday boardings on the central routes. The sample is the 100 boardings represented by the selected records.

Population parameter	Sample statistic
μ, mean fare paid across all central-route boardings that Monday	$\bar{x} = 3.2$ thousand COP, mean among the 100 boardings
σ, standard deviation of fares across all those boardings	$s = 0.8$ thousand COP, standard deviation among the 100 boardings
p, fraction of all those boardings that used a discounted fare	$\hat{p} = 35/100 = 0.35$, fraction among the 100 boardings



C1 – Compare what changes when routes are added

Quantity	Original scenario	Revised scenario	What the change tells us
Population	8,400 Monday boardings on central routes	12,000 Monday boardings on central and suburban routes	The population gains $12{,}000 - 8{,}400 = 3{,}600$ boardings. The day stays the same.
Sample	100 central-route boardings	100 boardings from the combined routes	The sample size stays at 100. The pool it is drawn from expands.
Mean	μ for central routes, $\bar{x} = 3.2$ thousand COP	μ for combined routes, $\bar{x} = 3.4$ thousand COP	$3.4 - 3.2 = +0.2$ thousand COP in $\bar{x}$. Whether μ changed is unknown.
Standard deviation	σ for central routes, $s = 0.8$ thousand COP	σ for combined routes, $s = 0.9$ thousand COP	$0.9 - 0.8 = +0.1$ thousand COP in s. Whether σ changed is unknown.
Discounted-fare share	p for central routes, $\hat{p} = 35/100 = 0.35$	p for combined routes, $\hat{p} = 30/100 = 0.30$	$0.30 - 0.35 = -0.05$ in $\hat{p}$ (-5 percentage points). Whether p changed is unknown.



1.4 A coffee association's export shipments (C1)

A coffee exporters' association wants to describe all 900 coffee shipments in its April export records. In a random sample of 40 shipments, the mean shipment weight is 18.4 metric tonnes and the sample standard deviation is 4.0 metric tonnes. Twelve of the sampled shipments departed on schedule.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 A second analyst repeats the study with a fresh random draw of 40 shipments from the association's April records. The mean shipment weight is 19.1 metric tonnes and the standard deviation is 3.6 metric tonnes. Twelve of these shipments departed on schedule. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The population is all 900 coffee shipments in the association's April export records. The sample is the 40 selected shipments.

Population parameter	Sample statistic
μ, mean weight across all 900 April shipments	$\bar{x} = 18.4$ metric tonnes, mean among the 40 shipments
σ, standard deviation of weight across all 900 April shipments	$s = 4.0$ metric tonnes, standard deviation among the 40 shipments
p, fraction of all 900 April shipments that departed on schedule	$\hat{p} = 12/40 = 0.30$, fraction among the 40 shipments



C1 – Compare what changes across samples

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All 900 shipments in the April records	Same 900 shipments	μ, σ and p stay fixed.
Sample	First draw of 40 shipments	Second draw of 40 shipments	The sample size stays the same. The draw is new.
Mean	$\bar{x} = 18.4$ metric tonnes	$\bar{x} = 19.1$ metric tonnes	$19.1 - 18.4 = +0.7$ metric tonnes in $\bar{x}$.
Standard deviation	$s = 4.0$ metric tonnes	$s = 3.6$ metric tonnes	$3.6 - 4.0 = -0.4$ metric tonnes in s.
On-schedule share	$\hat{p} = 12/40 = 0.30$	$\hat{p} = 12/40 = 0.30$	$0.30 - 0.30 = 0$ in $\hat{p}$ (0 percentage points).



1.5 A credit union's loan review (C1)

A credit union wants to describe all 800 of its active microenterprise working-capital loans on June 30. In a random sample of 50 loans, the mean outstanding balance is 8.0 million COP and the sample standard deviation is 2.4 million COP. Ten of the sampled loans have an overdue installment on that date.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 The lending committee asks the analyst to include active microenterprise equipment loans in the June 30 review. The two loan categories together contain 1,000 loans. In a random sample of 50 from the combined list, the mean outstanding balance is 8.8 million COP and the standard deviation is 2.9 million COP. Eight sampled loans have an overdue installment on that date. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The population is all 800 active microenterprise working-capital loans on June 30. The sample is the 50 selected loans.

Population parameter	Sample statistic
μ, mean outstanding balance across all 800 loans	$\bar{x} = 8.0$ million COP, mean among the 50 loans
σ, standard deviation of outstanding balances across all 800 loans	$s = 2.4$ million COP, standard deviation among the 50 loans
p, fraction of all 800 loans with an overdue installment on June 30	$\hat{p} = 10/50 = 0.20$, fraction among the 50 loans



C1 – Compare what changes when a loan category is added

Quantity	Original scenario	Revised scenario	What the change tells us
Population	800 active working-capital loans	1,000 active working-capital and equipment loans	The population gains $1{,}000 - 800 = 200$ loans. The review date stays the same.
Sample	50 working-capital loans	50 loans from the combined list	The sample size stays at 50. The pool it is drawn from expands.
Mean	μ for working-capital loans, $\bar{x} = 8.0$ million COP	μ for both loan categories, $\bar{x} = 8.8$ million COP	$8.8 - 8.0 = +0.8$ million COP in $\bar{x}$. Whether μ changed is unknown.
Standard deviation	σ for working-capital loans, $s = 2.4$ million COP	σ for both loan categories, $s = 2.9$ million COP	$2.9 - 2.4 = +0.5$ million COP in s. Whether σ changed is unknown.
Overdue share	p for working-capital loans, $\hat{p} = 10/50 = 0.20$	p for both loan categories, $\hat{p} = 8/50 = 0.16$	$0.16 - 0.20 = -0.04$ in $\hat{p}$ (-4 percentage points). Whether p changed is unknown.

